

How Much Telemetry Is Enough?

Predicting High-Skill StarCraft II Players with Four Versus Fifteen Behavioral Sensors

Extended abstract classroom demonstration · Merit Point Academy · 20 August 2026

Takeaway — Four interpretable gameplay sensors retained most of the discrimination achieved by all fifteen telemetry variables. Across identical five-fold splits, the area under the receiver operating characteristic curve (ROC AUC) rose from 0.861 to 0.879 with the larger sensor set—a modest 0.018 gain that motivates a controlled sensor-ablation study rather than a causal claim.

Background

Game telemetry turns play into sequences of measurable actions, creating opportunities for player modeling and adaptive game systems. For a classroom agent project, however, using every available signal can make the model harder to explain. We asked how much predictive performance is lost when a classifier for higher-ranked StarCraft II players uses four interpretable behavioral sensors rather than all fifteen telemetry variables in an open dataset. The study is a bridge between a browser-game activity, where an agent sees four numeric sensors, and a reproducible analysis of real gameplay records.

Method

We used the SkillCraft1 Master Table Dataset from the University of California, Irvine (UCI) Machine Learning Repository under the Creative Commons Attribution 4.0 license (CC BY 4.0). It contains 3,395 player records. We operationalized “high skill” as LeagueIndex 5–8 (Diamond through Professional; $n = 1,517$) and the comparison class as LeagueIndex 1–4 ($n = 1,878$). The four-sensor model used actions per minute (APM), selection by hotkeys, action latency, and number of perception–action cycles (PACs). The full model used all fifteen available behavioral telemetry features. Each condition used z-scored inputs and the same class-balanced logistic-regression specification. We evaluated both conditions with identical five-fold stratified cross-validation splits, shuffled with `random_state = 42`. Accuracy, balanced accuracy, and ROC AUC were calculated for every held-out fold; reported spreads are standard deviations (SDs) across those five folds. A ROC curve plots the true-positive rate against the false-positive rate over decision thresholds; its AUC summarizes ranking discrimination (0.5 = chance, 1.0 = perfect).

Results

The four-sensor model achieved accuracy 0.776 ± 0.013 , balanced accuracy 0.776 ± 0.013 , and ROC AUC 0.861 ± 0.011 . The fifteen-sensor model achieved accuracy 0.789 ± 0.012 , balanced accuracy 0.789 ± 0.010 , and ROC AUC 0.879 ± 0.012 . Thus, adding eleven telemetry variables was associated with a 0.018 increase in mean ROC AUC and a 0.012 increase in mean accuracy. The result shows that the compact sensor set retained most of the full model’s discrimination on these folds, while the larger set produced a modest improvement. It does not establish that any sensor causes a player to become more skilled.

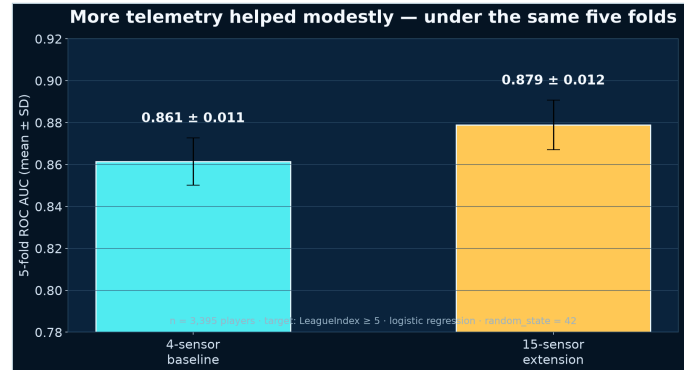


Figure 1. Mean ROC AUC $\pm$ standard deviation across the same five stratified folds. The 15-sensor extension exceeded the 4-sensor baseline by 0.018 ROC AUC.

Limitations

LeagueIndex is an operational label based on competitive rank, not a complete definition of expertise. The analysis uses one historical game dataset and one linear classifier; it does not test other games, time periods, model families, or prospective players. Repeated records, unmeasured context, and telemetry construction could affect the estimates. Cross-validation reduces dependence on one train/test split but does not provide external validation. The observed metric differences are descriptive; no inferential test or confidence interval was specified in the evidence receipt.

Next Experiment

Run a preregistered sensor-ablation ladder with the same five folds: begin with APM alone, then add SelectByHotkeys, ActionLatency, and NumberOfPACs one at a time, followed by the remaining sensors in documented groups. Record fold-level ROC AUC, balanced accuracy, calibration, and inference time for every step. This controlled sequence would show which additions provide stable incremental value and whether the four-sensor model offers a useful accuracy–interpretability trade-off.

Open Artifact

Analysis code, exact fold-level outputs, aggregate results, prompt, and figure are available in the workshop’s [open-data demonstration package](#). The raw dataset is retrieved from UCI at run time; the saved download uses the 256-bit Secure Hash Algorithm (SHA-256):
d6f4f88c2d077a0938cf774ff242df511c9a5801dcc66e4461ff

Reference

Blair, M., Thompson, J., Henrey, A., & Chen, B. (2013). *SkillCraft1 Master Table Dataset*. UCI Machine Learning Repository. <https://doi.org/10.24432/C5161N>. Licensed CC BY 4.0.

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